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The Kiel *RN* experiment: final porcine chromosome 15 mapping results

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Introduction

The post-mortem pH decline in Hampshire pigs does not differ from that found in other breeds at an early stage, but terminal pH shows a markedly lowered level in Hampshires. This phenomenon has been called 'Hampshire type meat' (MONIN and SELLIER 1985) or 'acid meat' (SELLIER 1990). Technological yield in cured cooked ham processing of Hampshire meat has been found to be significantly lowered. NAVEAU et al. (1985) created an artificial trait called Rendement Technologique NAPOLE, which is closely related to technological yield but more convenient to measure. NAVEAU (1986) found a bimodal distribution of this trait in two composite lines, with theoretical proportions of 50 and 33 % of their genes from the hampshire breed. From this finding he put forward the hypothesis of the existence of a single locus (the *RN* locus) with one dominant allele (RN^-) with an unfavourable effect on technological yield and one normal recessive allele (rn^+). LEROY et al. (1990) examined this hypothesis, using complex segregation analysis. As postulated by NAVEAU (1986), the dominant mode of inheritance was found to be the most likely one, when compared to a set of alternatives.

Measurements of the glycolytic potential (weighted sum of concentrations of lactate and its preliminary stages in muscle energy metabolism: glucose, glucose-6-phosphate, and, most important, glycogen) showed an increased level in Hampshires (SAYRE et al. 1963; MONIN and SELLIER 1985; MONIN et al. 1987; WASSMUTH and GLODEK 1992; FEDDERN 1994). Again, bimodal distributions were found in lines comprising certain amounts of Hampshire genes (FERNANDEZ et al. 1990). It is well accepted now that the RN^- allele affects both glycolytic potential and technological yield. The most important constituent of the glycolytic potential is glycogen. Muscle glycogen content is sharply increased in RN^- carriers, both when measured in biopsy samples from live animals and at different times after slaughter. Increased muscle glycogen content (and decreased protein content) is obviously the primary RN^- effect: the water-binding ability of glycogen is similar to protein (3.5 g water per g protein) in muscle and fresh meat (MONIN 1989). However, during cooking glycogen is denaturated, and a huge proportion of the water bound to it is set free.

Based on single-gene dominant inheritance, the *RN* locus was mapped to chromosome 15 by MILAN et al. (1995), when the phenotypic trait measured was Rendement Technologique NAPOLE. Another mapping experiment using terminal muscle glycogen content was conducted at the University of Kiel. Mapping results from a subset of all animals, which were part of this experiment, and from four chromosome 15 markers (RUDAT et al. 1995; LOOFT et al. submitted), confirmed the findings reported by MILAN et al. (1995). Recently, MILAN et al. (1996) presented a chromosome 15 map, comprising a set of 16 microsatellite markers as well as the *RN* locus. Furthermore, the closest marker, Sw936, to *RN* was physically mapped to 15q21–22 by DISC-PCR. At the same time, MARIANI et al. (1996) positioned *RN* between S0088 and the marker pair Sw936/Sw906 on chromosome 15, based on measurements of glycolytic potential and on 'residual glycogen', i.e. the unweighted sum

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of concentrations of glycogen, glucose, and glucose-6-phosphate. Although it is now well established that the position of the *RN* region is on chromosome 15, there is some contradiction between studies with the map details. These contradictions concern the position of Sw936 in relation to *RN* and Sw906. It is not clear whether Sw936 is nearly identical with the *RN* bound or with the Sw906 bound of the *RN*–Sw906 interval.

This paper presents the chromosome 15 mapping results from the Kiel experiment for the *RN* locus, giving complete data from all animals and seven different microsatellite markers. The focus is on map details of the *RN* region. Additional information is given on marker informativeness and differences in recombination rates between sexes.

Materials and methods

Animals

A previous experiment (FEDDERN 1994) revealed a bimodal distribution of the glycolytic potential in Hampshire × Piétrain and (Hampshire × Piétrain) × (Landrace × Large White) pigs from a commercial crossbreeding programme. The distribution of glycogen was unimodal, with a low mean value in Landrace × Large White crosses and again, but with a high mean value, in purebred Hampshires from the same breeding programme. These findings were interpreted as evidence for the existence of the RN^- allele in the Hampshire line and its absence in the Piétrain, Landrace and Large White lines used in this programme. Therefore 16 Hampshire × Piétrain crossbred boars were mated to Large White, Landrace, and Large White × Landrace sows to produce families for our mapping experiment. Linkage data comprised information from 15 boars, 61 sows (5, 12, and 44 of them Landrace, Large White, and Landrace × Large White, respectively), and 509 progenies, giving a total of 585 animals. The origin of pigs was the same as in the experiment described by FEDDERN (1994). All offspring were fattened at the Hohenschulen experimental piggery of the Christian-Alberchts University near Kiel, under the usual production conditions. The animals were slaughtered at around 110 kg liveweight in two different slaughterhouses, using the techniques of either electronarcosis or CO_2 stunning and exsanguination.

Muscle tissue samples

A slice from the musculus longissimus dorsi, between the 13th and 14th rib, was cut from each cooled carcass, about 24 h after slaughter. After transport to the institute's laboratory, eight cubes of approximately 0.5 cm side length were taken from each slice and shock frozen on dry ice. Samples were kept at −20 °C. The glycogen content was determined from approximately 0.5 g of muscle tissue, using the iodine binding method (DREILING et al. 1987). Each glycogen value is the average of two independent measurements from the same musculus longissimus dorsi slice. Measurements are expressed as μmol lactate equivalents per g wet muscle tissue.

Assignment of genotypes

Terminal muscle glycogen content was used as a phenotypic trait for genotype assignment. This quantity was preferred to glycolytic potential, for its simplicity. This is possible, due to a close non-linear relationship between glycolytic potential and glycogen content (Fig. 1; data from FEDDERN 1994). Fourteen boars were classified as heterozygous (RN^-/rn^+), because two types of extreme animals, with either normal or high glycogen content, were found within their progeny. Two boars which had only offspring with normal glycogen values were assumed to be not informative at the *RN* locus (likely to be homozygous rn^+/rn^+). One of them was excluded from the experiment, the other boar was used for marker typing only (*RN* genotypes of this family treated as unknown). Four hundred and

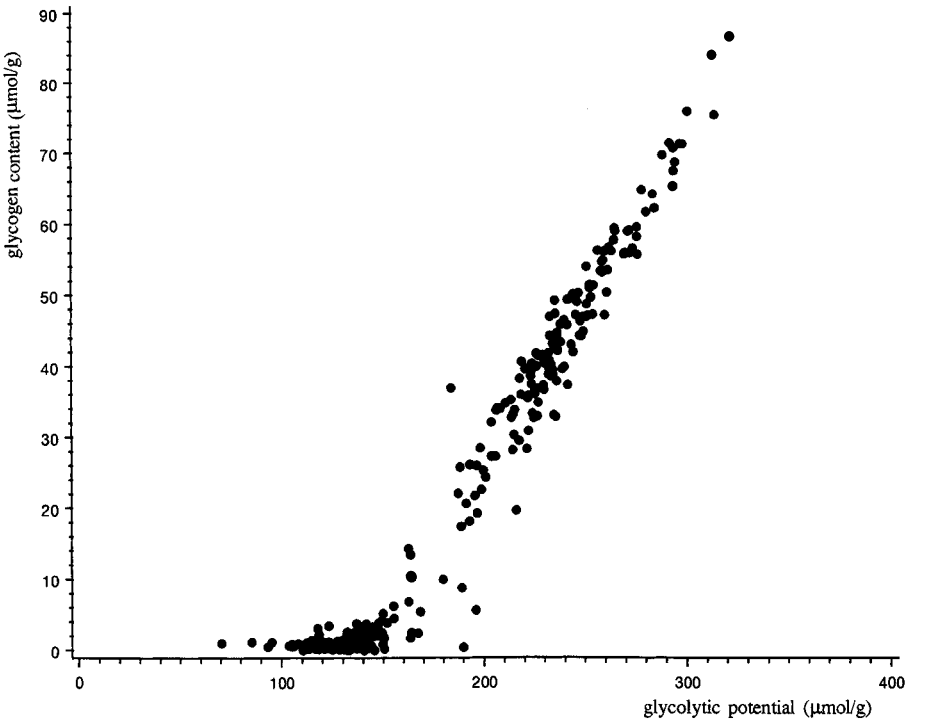


Fig. 1. Terminal glycogen content (glycose equivalents) in musculus longissimus dorsi samples versus glycolytic potential 24 h post mortem (data from FEDDERN 1994)

ninety-six progenies out of 509 were from heterozygous boars. Offspring with a low glycogen content, up to the 30 % percentile of the glycogen distribution (7 µmol/g), were classified as rn^+/rn^+, and those with a glycogen content above the 70 % percentile (100 µmol/g) as RN^-/rn^+. The genotype of all other offspring with glycogen content out of a medium region of uncertainty (7 µmol/g < glycogen content ⩽ 100 µmol/g) was treated as unknown (equivalent to a proportion of 40 % unknown genotypes among progenies from heterozygous sires). Based on the results found in FEDDERN (1994), all sows were treated as rn^+/rn^+.

Markers and linkage analysis

DNA extraction roughly followed the protocol of WANG et al. (1994). Seven different chromosome 15 microsatellite markers were selected: S0088 (ELLEGREN et al. 1994), Sw312, Sw936, Sw964, Sw919, Sw906, Sw120 (ROHRER et al. 1994). Polymerase chain reaction (PCR) amplification was performed in a volume of 15 µl, containing 15–25 ng genomic DNA, 1.5 mM $MgCl_2$, 20 mM $(NH_4)_2SO_4$, 75 mM Tris HCl (pH 9.0), 100 µM dNTP, 0.0625 U Goldstar Polymerase (Eurogentec), and 0.5 µM of each primer. For Sw919, the pH of Tris HCl was 9.5. The forward primer was labelled with fluorescein at the 5′-end. A GenAmp PCR system 9600 thermocycler (Perkin Elmer) was employed to conduct the following amplifications steps: 94 °C for 3 min, followed by a marker-specific number of cycles (Table 1), each cycle consisting of 20 s at 94 °C for denaturation, 30 s at a marker-specific temperature (Table 1) for annealing, and 30 s at 72 °C for elongation. Final extension lasted 10 min at 72 °C in all cases. PCR was carried out separately for each marker. Whenever possible, amplification products and internal size standards of two to three markers were

Table 1. **Annealing temperatures and number of PCR cycles for marker amplification**

	Marker						
	Sw919	Sw964	S0088	Sw120	Sw906	Sw936	Sw312
Annealing temperature (°C)	58	60	60	60	55	60	60
Number of PCR cycles	35	24	25	32	30	25	25

Table 2. **Five modes of genotype assignment for *RN* in progenies, based on individual muscle glycogen content (gly, in μmol/g). Results from RN1 are reported, variants RN2–RN5 were used for sensitivity analysis. Lower and upper bounds of the uncertainty region (unknown genotypes) are expressed in absolute values (column 3) as well as percentiles of the glycogen distribution (e.g. last column, P30 denotes 30 % percentile)**

Variant	Genotypes and corresponding intervals on glycogen scale			Proportion of unknown genotypes (%)	Percentile bounds of uncertainty region
	rn^+/rn^+	Unknown	rn^+/RN^-		
RN1	gly ⩽ 7	7 < gly ⩽ 100	gly > 100	40	P30 < gly ⩽ P70
RN2	gly ⩽ 15	15 < gly ⩽ 79	gly > 79	20	P40 < gly ⩽ P60
RN3	gly ⩽ 24	24 < gly ⩽ 67	gly > 67	10	P45 < gly ⩽ P55
RN4	gly ⩽ 34	34 < gly ⩽ 63	gly > 63	6	P47 < gly ⩽ P53
RN5	gly ⩽ 47	–	gly > 47	0	–

combined and mixed with loading buffer. All electrophoretic typing work was carried out with an Automated Laser Fluorescence detection system (A.L.F., Pharmacia) in 6 % denaturating polyacrylamide gels. Raw data obtained from this device were converted to genotypic data, applying the Fragment Manager (Pharmacia) and Automated Linkage Preprocessor programs (MANSFIELD et al. 1994). To check for mendelian inheritance and potential typing errors, GENPRUEF software was used (REINSCH, unpubl. data). Two-point and multipoint maximum-likelihood linkage analysis was carried out with CRIMAP 2.4 (GREEN et al. 1990), using options TWOPOINT, BUILD, and FLIPS. The conformity of male and female recombination rates was investigated by likelihood ratio tests. The differences between lod scores from sex-specific and average two-point analyses were converted to a likelihood ratio test statistic, by multiplication with $2\log_{10}$. Significance levels from a χ^2 distribution with one degree of freedom were computed with the PROBCHI function of the SAS package (SAS INSTITUTE INC. 1990).

Sensitivity analysis

The proportion of offspring with unknown genotypes depends on the length of the region of uncertainty (7 μmol/g < glycogen content ⩽ 100 μmol/g), where unambiguous assignment of genotype based on individual glycogen content is thought to be impossible. Of course the size of this region is somewhat arbitrary. In order to obtain information on how the mapping results are influenced by the proportion of progeny treated as unknown, multipoint analysis was repeated with four different definitions of this region, with proportions of unknown genotypes among progenies of 20 %, 10 %, 6 %, and 0 %, respectively. In all cases, the uncertainty region was centred at the median of the glycogen distribution. For the exact specification of intervals, see Table 2.

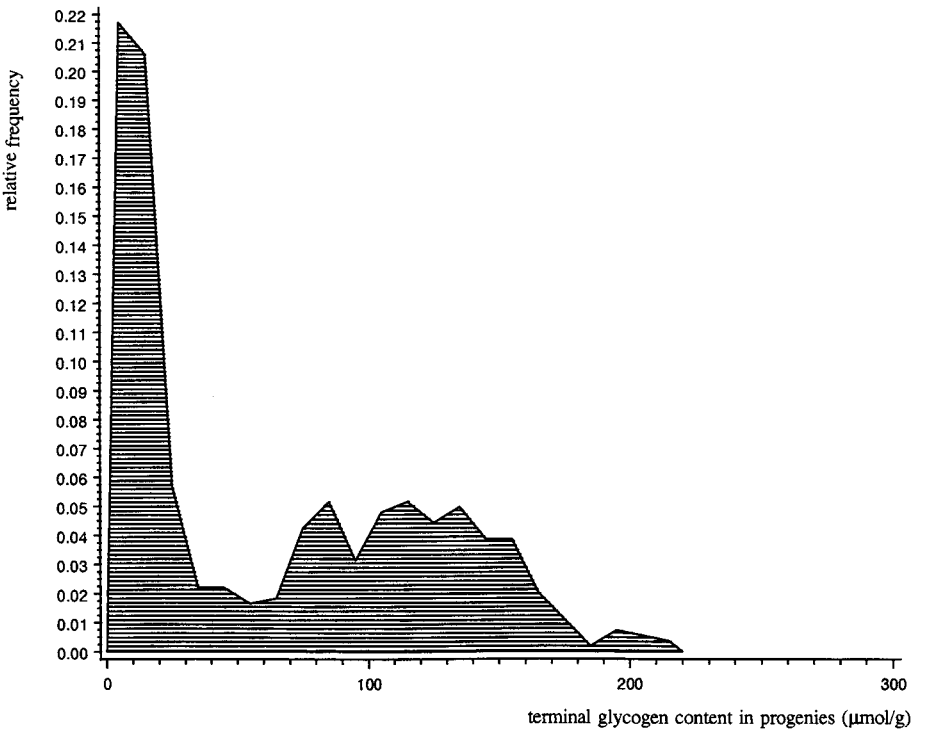


Fig. 2. Frequency distribution of terminal glycogen content in musculus longissimus dorsi samples from offspring

Results and discussion

Musculus longissimus dorsi glycogen content ranged up to 208 µmol/g and showed a clear bimodal distribution (Fig. 2): the lower part of the distribution located left from the median (47 µmol/g) exhibited a sharply spiked shape, while the upper part was very broad and flat. This indicates that glycogen content in homozygotes (rn^+/rn^+) has not only a lower mean level but, additionally, is markedly less variable than in heterozygotes (rn^+/RN^-). This aspect of terminal muscle glycogen content, has not, to the authors' knowledge, been previously reported in the literature. Factors such as environmental temperature, or amount of time elapsed from slaughter to freezing muscle tissue samples in the laboratory, are not fully under experimental control, and are likely to be sources of variation in glycogen content.

The informativeness of marker loci and *RN* is summarized in Table 3. The minimum number of different alleles was four (S0088) and the maximum 14 (Sw919). The degree of heterozygosity varied from 0.47 to 1.00 in males and from 0.71 to 0.88 in females. With few exceptions, all alleles could be found in Hampshire × Piétrain boars, as well as in Landrace × Large White sows. Since porcine microsatellite markers were initially developed and mapped in crosses between genetically very distant genotypes (e.g. Chinese Meishan and European Large White (ROHRER et al. 1994) or Large White and the Wild Boar (ELLEGREN et al. 1994)) it is doubtful whether the heterozygosity of these markers is sufficient for application in crosses between genetically more narrow breeds. However, these markers performed well in the families derived from commercially selected lines.

The *RN* gene was positioned in the centre of an interval flanked by Sw120 and the marker

Table 3. **Informativeness of marker loci and *RN***

Locus	Different alleles observed (no.)	Heterozygosity in boars	Heterozygosity in sows	Informative meioses (phase unknown) (no.)	Range of fragment sizes
Sw919	14	1.00	0.875	802	85–151
Sw964	7	0.800	0.750	593	219–243
S0088	4	0.467	0.707	472	145–161
Sw120	7	1.00	0.714	754	143–157
Sw906	6	0.667	0.722	614	158–186
Sw936	8	0.933	0.707	675	90–112
Sw312	9	0.867	0.881	691	126–142
RN	2	0.875	0.000	300	–

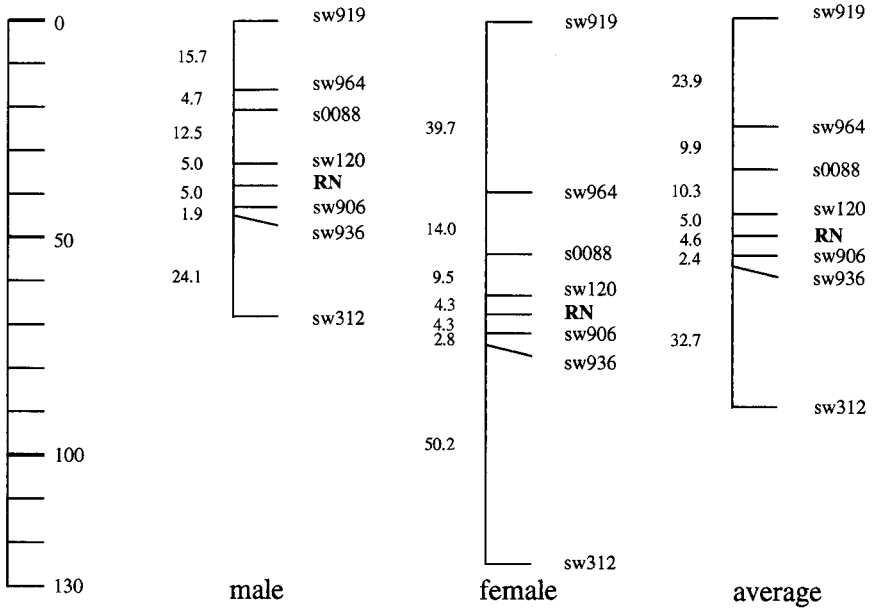


Fig. 3. Map of chromosome 15 (Kosambi centimorgan)

pair Sw906/Sw936 (Fig. 3). The length of this interval in males was estimated as 10 Kosambi cM. Male recombination rates between *RN* and the three closest markers, Sw120, Sw906, and Sw936, were 0.05 in all cases when estimated via two-point maximum likelihood analysis (Table 4), but Sw906 was mapped closer to *RN* in multipoint runs. Male multipoint map details of the *RN* region, from this experiment, are compared with the findings of MARIANI et al. (1996) and MILAN et al. (1996) in Figure 4. If Sw936 is not considered, this map is in good agreement with that illustrated by MILAN et al. (1996), who reported a male distance of 7 cM both to Sw120 and Sw906 (Fig. 4). However, there is a contradiction with Sw936,

Table 4. **Recombination rates between markers and *RN* and corresponding lod scores from two-point analysis**

	Marker locus						
	Sw919	Sw964	S0088	Sw120	Sw906	Sw936	Sw312
Recombination rate	0.27	0.18	0.20	0.05	0.05	0.05	0.27
lod score	8.03	12.87	6.98	52.09	29.30	42.53	5.85

which is reported to be nearly identical to *RN* in the study of MILAN et al. (1996). The multipoint estimate of 4.8 cM for the male length of the *RN*–Sw906 interval presented by MARIANI et al. (1996), coincides with the result from this study of 5.0 cM.

Moreover, the location of Sw936 on the map of the latter group is quite similar to the findings here (Fig. 4). ROHRER et al. (1994) gave a small distance for markers Sw936/ Sw906, which is similar to the findings found in this study and in MARIANI et al. (1996). For completeness, it should be mentioned that the locus order remains unchanged, when the markers in this study are analysed without *RN* data. To summarize, there is good agreement between studies on the localization of the *RN* locus on chromosome 15 in relation to Sw906, in spite of contradictions about the exact organization of loci in the *RN* region. The marker Sw936 seems to be more remote from *RN* than estimated by MILAN et al. (1996). Additional markers, located closer to *RN*, would be helpful in the search for clones with DNA fragments out of the *RN* region in YAC libraries.

Total male map length of the chromosome 15 region flanked by our markers was 68.8 cM, compared to a female map length of 124.7 cM, i.e. The female map length was 1.8 times that of the male map in this particular region of the porcine genome. ELLEGREN et al. (1994) reported an average porcine male to female recombination ratio of 1 : 1.4 for the entire

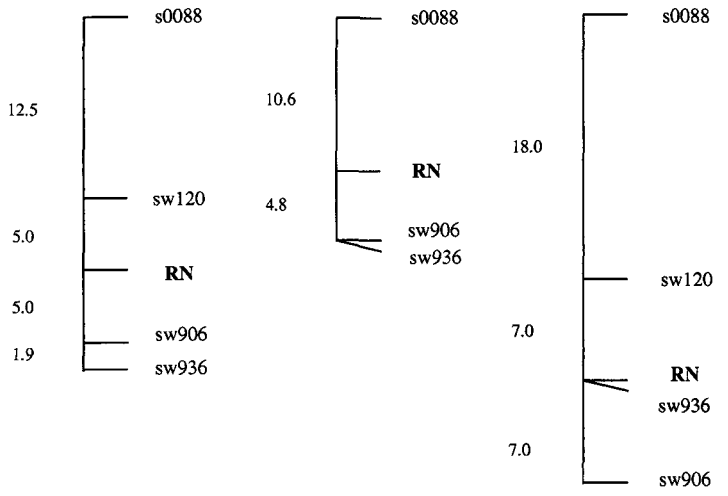


Fig. 4. Comparison of male multipoint map details in the *RN* region from three studies

Table 5. **Sex-specific and mean recombination rates, corresponding lod scores and likelihood ratio tests for sex differences in recombination rates at six adjacent chromosome 15 intervals**

Interval	Recombination rates			lod scores		Test statistic χ^2	Significance level
	Female	Male	Sex Mean	specific	Mean		
Sw919–Sw964	0.33	0.15	0.23	24.51	21.44	18.28	0.0000*
Sw964–S0088	0.15	0.05	0.12	31.41	29.94	6.770	0.0093*
S0088–Sw120	0.10	0.12	0.11	46.31	46.23	0.3684	0.5439
Sw120–Sw906	0.09	0.08	0.08	104.69	104.61	0.0800	0.7773
Sw906–Sw936	0.03	0.02	0.02	95.77	95.66	0.5066	0.4776
Sw936–Sw312	0.38	0.22	0.28	18.98	16.27	12.48	0.0004*

* Significant at $P < 0.01$.

genome, in contrast to 1 : 1.7 in humans. The difference between the male and female recombination rate was significant for three intervals bounded by marker pairs Sw919–Sw964, Sw964–S0088, and Sw936–Sw312 (Table 5). No significant difference was found in the neighbourhood region of the *RN* gene (S0088–Sw936). ELLEGREN et al. (1994) discussed the possibility of variation in recombination rates between domestic breeds and their wild ancestors, but found no indications for such differences in their experimental data. If variation between breeds, recombination rates exists in pigs, then the difference between sexes in this investigation is confounded with such a breed difference between Piétrain × Hampshire and Landrace × Large White crossbreds.

Mapping results were insensitive to different variants of genotype assignment for progenies. Male and female map length, as well as distances between *RN* and the closest markers Sw120, Sw906, and Sw936, were nearly unaffected when the proportion of unknown genotypes in progenies was varied. There was a response in locus order and map distances only in the extreme case with no unknown genotypes at all. This is, of course, due to the occurrence of misclassifications. As a result, from sensitivity analysis, it can be concluded that the reported mapping results are highly reliable, in spite of some arbitrariness that is inherent in genotype assignment for offspring.

The fact that 14 of 16 of the crossbred boards initially used in matings were heterozygous, as well as the unimodal high-level distribution of glycolytic potential in purebred Hampshire observed by FEDDERN (1994), both suggest that the RN^- frequency is high in the Hampshire line used in the crossbreeding programme, from where these animals originated. Because of the dominant mode of inheritance in such a situation, the majority of the favourable recessive rn^+ alleles of the population is expected to be carried by heterozygotes. Markers could be used in combination with glycogen measurements from biopsy samples of live animals or slaughtered progenies, in order to select against RN^- carriers within families. As discussed by MILAN et al. (1996) in the context of selection within a synthetic line, a linkage disequilibrium between markers and *RN*, if it existed, could be exploited for the same purpose. Unfortunately, in this experiment, there is no information on the existence of such a linkage disequilibrium, since Hampshire × Piétrain crossbred boars were used, and all markers associated with RN^- are simply markers for the Hampshire origin of the unfavourable mutation. The existence or non-existence of a linkage disequilibrium should be examined in a separate study. It should be mentioned that the probability of a linkage disequilibrium, and its usefulness for breeding purposes, are not dependent on the distance between *RN* and the closest marker (SMITH and SMITH 1993). In any case, markers may be employed as a useful tool in breeding work up to the time when a DNA probe becomes available. Based on the homology of human chromosome 2q to the porcine chromosome 15 (RETTENBERGER

et al. 1995), promising candidate genes have been identified in previous publications (RUDAT et al. 1995; MARIANI et al. 1996; MILAN et al. 1996): the uridyl transferase 2 and the glucagon gene.

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Summary

A marker experiment with pigs from commercially selected lines is described. One important goal of the experiment was to map the porcine *RN* locus, a major gene responsible for lowered terminal pH and increased glycogen level in muscle tissue. Experimental families comprised 15 Piétrain × Hampshire boars, of which 14 were heterozygous at the *RN* locus, 61 homozygous rn^+/rn^+ sows (five Landrace, 12 Large White, and 44 Landrace × Large White crossbreds), and 509 progenies, 496 of them from heterozygous boars, in total 585 animals. Genotyping was done for seven chromosome 15 microsatellite loci: Sw919, Sw964, S0088, Sw120, Sw906, Sw936, and Sw312. Genotype assignment for the *RN* locus was based on musculus longissimus dorsi glycogen content. The heterozygosity of markers ranged from 0.47 to 1 in boars and from 0.71 to 0.88 in sows. *RN* was mapped to the centre of an interval, flanked by Sw120 and the marker pair Sw906/Sw936. Male multipoint distances between *RN* and Sw120, Sw906, and Sw936 were estimated as 5 cM, 5 cM, and 6.9 cM, respectively. Two-point recombination rates between these markers and RN were 0.05 in all cases, with corresponding lod scores of 52.09, 29.30, and 42.53. Concordances and disconcordances of mapping results in the *RN* region from different studies are discussed. The male map length of the chromosome region covered was 68.8 cM, in contrast to the female map length of 124.7 cM. Significant differences in recombination rates between sexes were found in intervals Sw919–Sw964, Sw964–S0088, and Sw936–Sw312, but not in the neighbourhood of the *RN* gene.

Zusammenfassung

Es wird ein Kartierungsexperiment mit Schweinen aus kommerziell genutzten Linien beschrieben. Vordringliches Ziel dieses Experimentes war es, das *RN*-Gen zu kartieren, das für einen erniedrigten End-pH-Wert und einen erhöhten Glykogengehalt im Musklegewebe verantwortlich ist. Die Daten stammen von 15 Hampshire × Piétrain Ebern, von denen 14 am *RN* Locus heterozygot waren, 61 homozygoten rn^+/rn^+ Sauen (5 Landrasse-, 12 Large White und 44 Landrasse × Large White Kreuzungssauen) sowie 509 Nachkommen, 496 davon von heterozygoten Ebern, insgesamt also 585 Tiere. Für sieben Mikrosatellitenmarker auf Chromosom 15 wurden die Tiere typisiert: Sw919, Sw964, S0088, Sw120, Sw906 und Sw312. Die Bestimmung der Genotypen am *RN*-locus erfolgte anhand des Glykogenghalts im Musculus longissimus dorsi. Der Heterozygotiegrad der Marker lag zwischen 0,47 und 1 bei Ebern und zwischen 0,71 und 0,88 bei den Sauen. Als Position des *RN*-Gens wurde die Mitte eines Intervalles bestimmt, der von dem Marker Sw120 sowie dem Markerpaar Sw906/Sw936 begrenzt wird. Für das männliche Geschlect wurden die Kartenabstände zwischen *RN* und Sw120, Sw906 und Sw936 in einer Mehrpunktanalyse auf 5 cM, 5 cM und 6,9 cM geschätzt. Paarweise Schätzungen der Rekombinationsraten zwischen diesen Markern und *RN* betrugen in allen Fällen 0,5, mit zugehörigen lod scores vol 52,09, 29,30 und 42,53. Übereinstimmungen und Diskrepanzen zwischen Kartierungsergebnissen verschiedener Arbeitsgruppen werden diskutiert. Die Kartenlänge des mit Markern abgedeckten Teils von Chromsom 15 betrug beim männlichen Geschlecht 68,8 cM, im Gegensatz zu einer Länge von 124,7 cM beim weiblichen Geschlecht. Signifikante Unterschiede in den Rekombinationsraten beider Geschlechter wurden in den Intervallen Sw919–Sw964, Sw964–S0088 und Sw936–Sw312 gefunden, nicht jedoch in der Umgebung des *RN*-Genes.

References

DREILING, C. E.; BROWN, D. E.; CASALE, L.; KELLY, L., 1987: Muscle glycogen: comparison of iodine binding and enzyme digestion assays and application to meat samples. Meat Sci. **20**: 167–177.

ELLEGREN, H.; CHOWDHARY, B. P.; JOHANSSON, M.; MARKLUND, L.; FREDHOLM. M.; GUSTAVSSON, I.; ANDERSSON, L., 1994: A primary linkage map of the porcine genome reveals a low rate of genetic recombination. Genetics **137**: 1089–1100.

FEDDERN, E., 1994: Verlauf der postmortalen Glykogenolyse und Muskelstrukturmermale bei

Hampshire-Reinzuchttieren und verschiedenen Kreuzungskombinationen. Thesis, Christian-Albrechts-Universität zu Kiel.
FERNANDEZ, X.; NAVEAU, J.; TALMANT, A.; MONIN, G., 1990: Distribution du potential glycolytique dans une population porcine et relation avec le rendement 'Napole'. J. Rech. Porc. France **22:** 97–100.
GREEN, P.; FALLS, K.; CROOKS, S., 1990: CRIMAP Documentation. Washington University School of Medicine, Saint Louis.
JENSEN, P.; CRAIG, H. B.; ROBINSON, O. W., 1967: Phenotypic and genetic associations among carcass traits of swine. J. Anim. Sci. **26:** 1252–1260.
LOOFT, CHR.; REINSCH, N.; RUDAT, I.; KALM, E., 1996: Mapping the porcine *RN* gene to chromosome 15. (Submitted).
MANSFIELD, D. C.; BROWN, A. F.; GREEN, D. K.; CAROTHERS, A. D.; MORRIS, S. W.; EVANS, H. J.; WRIGHT, A. F., 1994: Automation of genetic linkage analysis using fluorescent microsatellite markers. Genomics **24:** 225–233.
MARIANI, P.; LUNDSTRÖM, K.; GUSTAFSSON, U.; ENFÄLT, A.-C.; JUNEJA, R. K.; ANDERSSON, L., 1996: A major locus (*RN*) affecting muscle glycogen content is located on pig chromosome 15. Mamm. Genome. **7:** 52–54.
MILAN, D.; LE ROY, P.; WOLOZYN, N.; CARITEZ, J. C.; ELSEN, J. M.; SELLIER, P.; GELLIN, J., 1995: The *RN* locus for meat quality maps to pig chromosome 15. Genet. Sel. Evol. **27:** 195–199.
MILAN, D.; WOLOSZYN, N.; YERLE, M.; LE ROY, P.; BONNET, M.; RIQUET, J.; LAHBIB-MANSAIS, Y.; CARITEZ, J.-C.; ROBIC, A.; SELLIER, P.; ELSEN, J.-M.; GELLING, J., 1996 Accurate mapping of the 'acid meat' *RN* gene on genetic and physical maps of pig chromosome 15. Mamm. Genome. **7:** 47–51.
MONIN, G., 1989: Genetic effects on technological qualities of pig meat. In: 40th Annual Meeting on the EAAP, Dublin,, paper 3.1.
MONIN, G.; SELLIER, P., 1985: Pork of low technological quality with a normal rate of muscle pH fall in the immediate post-mortem period: the case of the Hampshire breed. Meat Sci. **13:** 49–63.
MONIN, G.; MEJENES-QUIANO, A.; TALMANT, A.; SELLIER, P., 1987: Influence of breed and muscle metabolic type on muscle glycolytic potential and meat pH in pigs. Meat Sci. **20:** 149–158.
NAVEAU, J., 1986: Contribution a l'étude du déterminisme génétique de la qualité de viande porcine. Héritabilité du rendement technologique NAPOLE. J. Rech. Porc. France **18:** 265–276.
NAVEAU, J.; POMMERET, P.; LECHAUX, P., 1985: Proposition d'une méthode de mesure du rendement technologique: 'la méthode NAPOLE'. Techni-Porc **8:** 7–13.
RETTENBERGER, G.; KLETT, C.; ZECHNER, U.; KUNZ, J.; VOGEL,W.; HAMMEISTER, H., 1995: Visualization of the conservation of synteny between pigs and humans by heterologous chromosomal painting. Genomics **26:** 372–378.
ROHRER, G. A.; ALEXANDER, L. J.; KEELE, J. W.; SMITH, T. P.; BEATTIE, C. W., 1994: A microsatellite linkage map of the porcine genome. Genetics **136:** 231–245.
LE ROY, P.; NAVEAU, J.; ELSEN, J. M.; SELLIER, P., 1990: Evidence for a new major gene influencing meat quality in pigs. Genet. Res, Cambridge **55:** 33–40.
RUDAT, I.; LOOFT, CHR, REINSCH, N.; KALM, E., 1995: The Kiel *RN* project—mapping a major locus for meat quality. In: 46th Annual Meeting of the EAAP in Prague, Commission on Genetics, Session 5.
SAS INSTITUTE INC., 1990: SAS Language Reference, Version 6, 1st Ed. SAS Institute Inc., Cary, North Carolina.
SAYRE, R. N.; BRISKEY, E. J.; HOEKSTRA, W. G., 1963: Comparison of muscle characteristics and post mortem glycolysis in three breeds of swine. J. Anim. Sci. **23:** 1012–1020.
SELLIER, P., 1990: Genetic control of meat quality. In: 41st Annual Meeting of the EAAP, Toulouse, paper 3a.3.
SMITH, C.; SMITH, D. B., 1993: The need for close linkages for marker assisted selection for economic merit in livestock. Anim. Breed. Abstr. **61:** 197–204.
WANG, L.; HIRAYASU, K.; ISHIZAWA, M.; KOBOYASHI, Y., 1994: Purification of genomic DNA from human whole blood by isopropanol-fractionation with concentrated NaI and SDS. Nucl. Acid Res, **22:** 1774–1775.
WASSMUTH, R.; GLODEK, P., 1992: Einfluß des Hampshirefaktors und der Standzeit auf das glykolytische Potential und die Fleischbeschaffenheit bei Schweinen. Fleischwirtschaft **72:** 1299–1302.

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